Senad-Leandro Lemes Galera

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Education

Technical University of Munich (TUM) — Munich, Germany
Master of Science in Computer Science — *Apr. 2024 – Present*

- Student Club: TUMuchData
- Achievements: Estimathon 2nd place; HackaTUM 6th place

Technical University of Darmstadt (TUD) — Darmstadt, Germany
Bachelor of Science in Computer Science — *Oct. 2020 – Mar. 2024*

- GPA: 1.7
- Thesis: "Evaluation of NVMe-oF for disaggregated storage." (Grade: 1.3; Prof. Dr. Carsten Binnig)

Technical Skills

Languages: C/C++, C#, Java, Python, SQL, JavaScript, TypeScript, Scala, Rust
Systems & Hardware: CUDA, RDMA, NVMe-oF, Unified Virtual Memory (UVM), NVIDIA Open GPU Kernel Modules, RDBMS, ScaleStore, PCIe, NVLink, GraceHopper, Blackwell, YCSB, SSB, TPC-H
Frameworks & Tools: .NET, PyTorch, TensorFlow, JUnit, Docker, Git, Azure DevOps
Methodologies: CI/CD, DevOps, TDD, OOP, Functional Programming, Agile, Scrum

Experience

Software Engineer — Apr. 2026 – Present
Lautersee GmbH — *Munich, Germany*

- Develop full-stack C#/.NET applications focused on data archiving and VAT-related processes.
- Built a testing environment from scratch and introduced automated testing.
- Created CI pipelines using Azure DevOps, increasing test coverage from approximately 15% to 64%.

Student Management — Jun. 2024 – Dec. 2024
Technical University of Munich (TUM) — *Munich, Germany*

- Evaluated and processed credit transfer applications, ensuring academic alignment with faculty standards.
- Audited and restructured the internal wiki to improve knowledge transfer and process documentation.

Student Assistant — Oct. 2023 – Jan. 2024
PLCM, Technical University of Darmstadt (TUD) — *Darmstadt, Germany*

- Reviewed and documented previous research on gamification in Manufacturing Execution Systems (MES).
- Modeled the developed software, summarized key findings, and designed and implemented a chatbot.
- Supported students throughout their internships.

Teaching Assistant – Digitaltechnik & Rechnerorganisation — Oct. 2021 – Oct. 2023
ENCRYPTO / ESA, Technical University of Darmstadt (TUD) — *Darmstadt, Germany*

- Led one to two weekly exercise sessions, explaining lecture concepts in greater depth.
- Assessed students' eligibility for the final exam, created exam questions, and graded exams.

Software Engineer — May 2020 – May 2022
Software AG — *Darmstadt, Germany*

- Maintained and expanded test environments for web and desktop applications.
- Developed an automated reporting tool to analyze and format test results and distribute reports by email.
- Contributed to the migration of NaturalONE from desktop to web, focusing on backend development.

Research Projects

Stabilizing UVM Performance | *arXiv* | *Guided Research, TUM* 2026
Advisors: Michail Georgoulakis Misegiannis, Ferdinand Gruber, Prof. Dr. Jana Giceva

- Developed a memory-management framework for GPU workloads oversubscribing device memory, implemented directly in NVIDIA's open-source UVM driver. Designed counter-driven hot-region promotion and proactive cold-region reclamation to dynamically manage data placement between CPU and GPU memory. Achieved up to 10.6× speedup over standard UVM across multiple GPU architectures and workloads.

ShapeDream: 3D Point Cloud Completion | *GitHub* 2026

- Developed a diffusion-based framework for reconstructing complete 3D shapes from partial point-cloud observations. Conditioned a pre-trained MVDream model using PointNet++ features and LoRA fine-tuning to generate consistent multi-view images, which were reconstructed into 3D meshes using SnapGTR. Outperformed ShapeFormer across all evaluated object categories while requiring substantially less task-specific training data.

Publications

L. Arnemann, **S.-L. Lemes Galera**, S. Winter, and B. Schleich, "Gamification of Resource Consumption Monitoring of Products and Machines: A Cross-Platform and User-Friendly Approach," *Procedia CIRP*, vol. 122, pp. 569–574, 2024.

Community Engagement

Volunteer Tutor Jan. 2025 – Present
Grundschule Schwanthalerstraße *Munich, Germany*

- Every wednesday I tutor pupils with a migration background as a *Schülerpate*, supporting them with schoolwork and extracurricular activities.